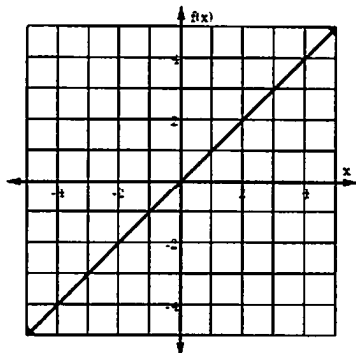


PARENT FUNCTIONS *Graphic Organizer*

A **function family** is a group of functions with similar characteristics. The **parent function** is the simplest of the functions in a family. You must be able to recognize each of these functions along with their characteristics.

LINEAR

Parent Function: $y = x$



Domain: $\mathbb{R}$

Range: $\mathbb{R}$

x-intercept(s): $(0,0)$

y-intercept: $(0,0)$

Extrema: None

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): None

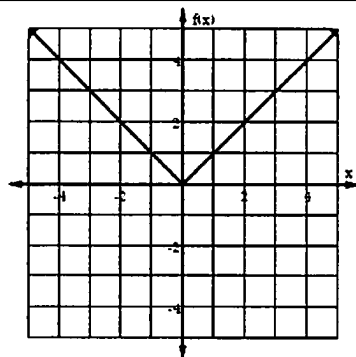
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$; As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Discontinuities? None

Even or Odd? odd

ABSOLUTE VALUE

Parent Function: $y = |x|$



Domain: $\mathbb{R}$

Range: $\{y | y \geq 0\}$

x-intercept(s): $(0,0)$

y-intercept: $(0,0)$

Extrema: $(0,0)$ - Absolute Minimum

Increasing Interval(s): $(0, \infty)$

Decreasing Interval(s): $(-\infty, 0)$

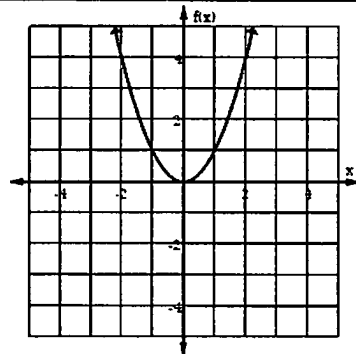
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$; As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$.

Discontinuities? None

Even or Odd? Even

QUADRATIC

Parent Function: $y = x^2$



Domain: $\mathbb{R}$

Range: $\{y | y \geq 0\}$

x-intercept(s): $(0,0)$

y-intercept: $(0,0)$

Extrema: $(0,0)$ - Absolute Minimum

Increasing Interval(s): $(0, \infty)$

Decreasing Interval(s): $(-\infty, 0)$

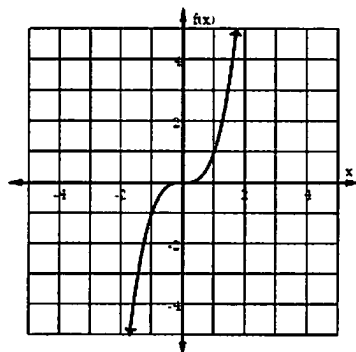
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$; As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$.

Discontinuities? None

Even or Odd? Even

CUBIC

Parent Function: $y = x^3$



Domain: $\mathbb{R}$

Range: $\mathbb{R}$

x-intercept(s): $(0,0)$

y-intercept: $(0,0)$

Extrema: None

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): None

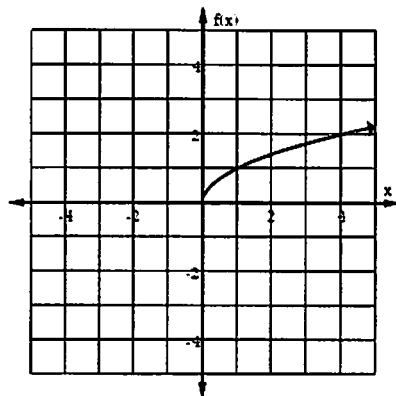
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$; As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

Discontinuities? None

Even or Odd? odd

SQUARE ROOT

Parent Function: $y = \sqrt{x}$



Domain: $\{x | x \geq 0\}$

Range: $\{y | y \geq 0\}$

x-intercept(s): $(0,0)$

y-intercept: $(0,0)$

Extrema: $(0,0)$ - Absolute Minimum

Increasing Interval(s): $[0, \infty)$

Decreasing Interval(s): None

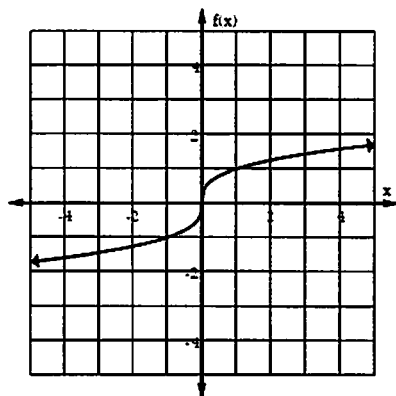
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$; As $x \rightarrow 0$, $f(x) \rightarrow 0$.

Discontinuities? None

Even or Odd? Neither

CUBEROOT

Parent Function: $y = \sqrt[3]{x}$



Domain: $\mathbb{R}$

Range: $\mathbb{R}$

x-intercept(s): $(0,0)$

y-intercept: $(0,0)$

Extrema: None

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): None

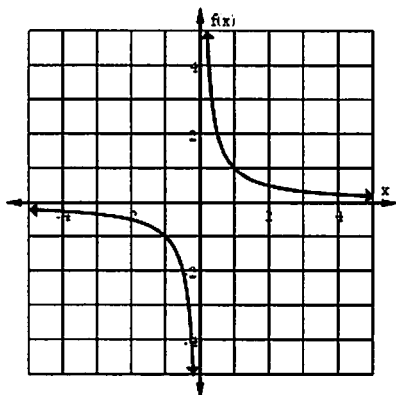
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$; As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

Discontinuities? None

Even or Odd? odd

RECIPROCAL

Parent Function: $y = \frac{1}{x}$



Domain: $\{x | x \neq 0\}$

Range: $\{y | y \neq 0\}$

x-intercept(s): None

y-intercept: None

Extrema: None

Increasing Interval(s): None

Decreasing Interval(s): $(-\infty, 0)$
 $(0, \infty)$

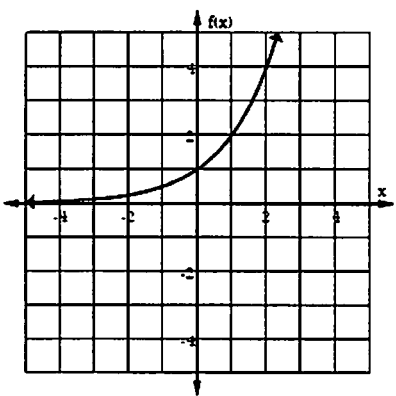
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow 0$; As $x \rightarrow -\infty$, $f(x) \rightarrow 0$.

Discontinuities? $x=0$;
infinite

Even or Odd? odd

EXPONENTIAL

Parent Function: $y = 2^x$



Domain: $\mathbb{R}$

Range: $\{y | y > 0\}$

x-intercept(s): None

y-intercept: $(0,1)$

Extrema: None

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): None

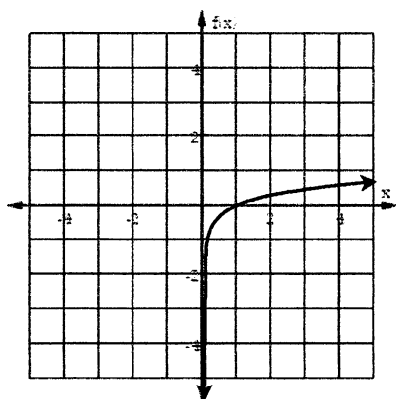
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$; As $x \rightarrow -\infty$, $f(x) \rightarrow 0$

Discontinuities? None

Even or Odd? Neither

LOGARITHMIC

Parent Function: $y = \log x$



Domain: $\{x | x > 0\}$

Range: $\mathbb{R}$

x-intercept(s): $(1, 0)$

y-intercept: None

Extrema: None

Increasing Interval(s): $(0, \infty)$

Decreasing Interval(s): None

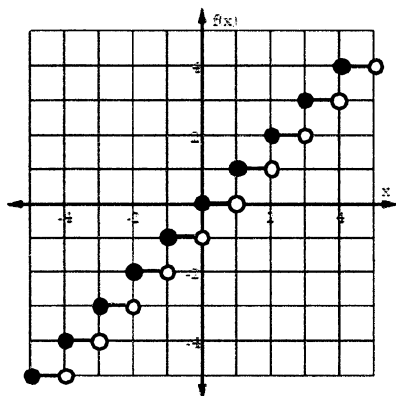
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$; As $x \rightarrow 0$, $f(x) \rightarrow -\infty$.

Discontinuities? None

Even or Odd? Neither

GREATEST INTEGER

Parent Function: $y = \lfloor x \rfloor$



Domain: $\mathbb{R}$

Range: $\{y | y \in \mathbb{Z}\}$
(All integers)

x-intercept(s): $(0, 0)$

y-intercept: $(0, 0)$

Extrema: None

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): None

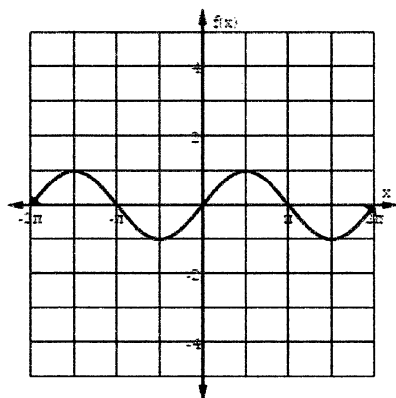
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$; As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

Discontinuities? jump;
every integer

Even or Odd? Neither

SINE

Parent Function: $y = \sin x$



Domain: $\mathbb{R}$

Range: $\{y | -1 \leq y \leq 1\}$

x-intercept(s): $(0, 0) + k\pi$

y-intercept: $(0, 0)$

Extrema: $(\frac{k\pi}{2}, 1)$ - Absolute Maximum $(-\frac{k\pi}{2}, -1)$ - Absolute Minimum

Increasing Interval(s): $(-\frac{\pi}{2}, \frac{\pi}{2}) + 2k\pi$

Decreasing Interval(s): $(\frac{\pi}{2}, \frac{3\pi}{2}) + 2k\pi$

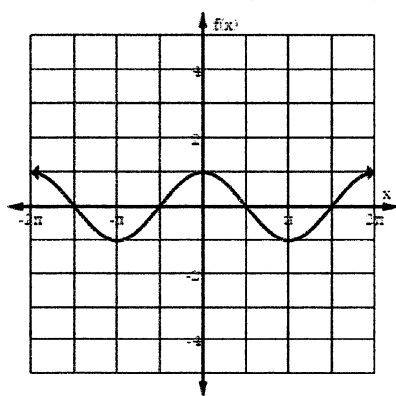
End Behavior: Does Not Exist

Discontinuities? None

Even or Odd? odd

COSINE

Parent Function: $y = \cos x$



Domain: $\mathbb{R}$

Range: $\{y | -1 \leq y \leq 1\}$

x-intercept(s): $(\frac{\pi}{2}, 0) + k\pi$

y-intercept: $(0, 1)$

Extrema: $(2k\pi, 1)$ - Absolute Maximum $(-k\pi, -1)$ - Absolute Minimum

Increasing Interval(s): $(\pi, 2\pi) + 2k\pi$

Decreasing Interval(s): $(-2\pi, -\pi) + 2k\pi$

End Behavior: Does Not Exist

Discontinuities? None

Even or Odd? Even